

Qatar-2b

R-1610

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_190529 · created 2026-08-02T03:17:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458632.77991 +/- 0.00082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.065
Transit depth (Rp/Rs)^2	1.8e-05 +/- 0.0055 [%]
Orbital Inclination (inc)	88.77 +/- 1.73
Airmass coefficient 1 (a1)	1.3 +/- 0.88
Airmass coefficient 2 (a2)	0.11 +/- 0.49
Scatter in the residuals of the lightcurve fit is	68.07 %
Best Comparison Star	#3 - [84, 207]
Optimal Aperture	2.76
Optimal Annulus	3.25
Transit Duration (day)	0.0675 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.065 vs 0.16327 (deviation 2.51 σ)
Duration fit vs published	0.0675 d vs 0.0767 d (deviation 2.7 σ)
Mid-time O-C	0.9 min
Depth SNR	0.0
Residual scatter	68.07 %

Light curve

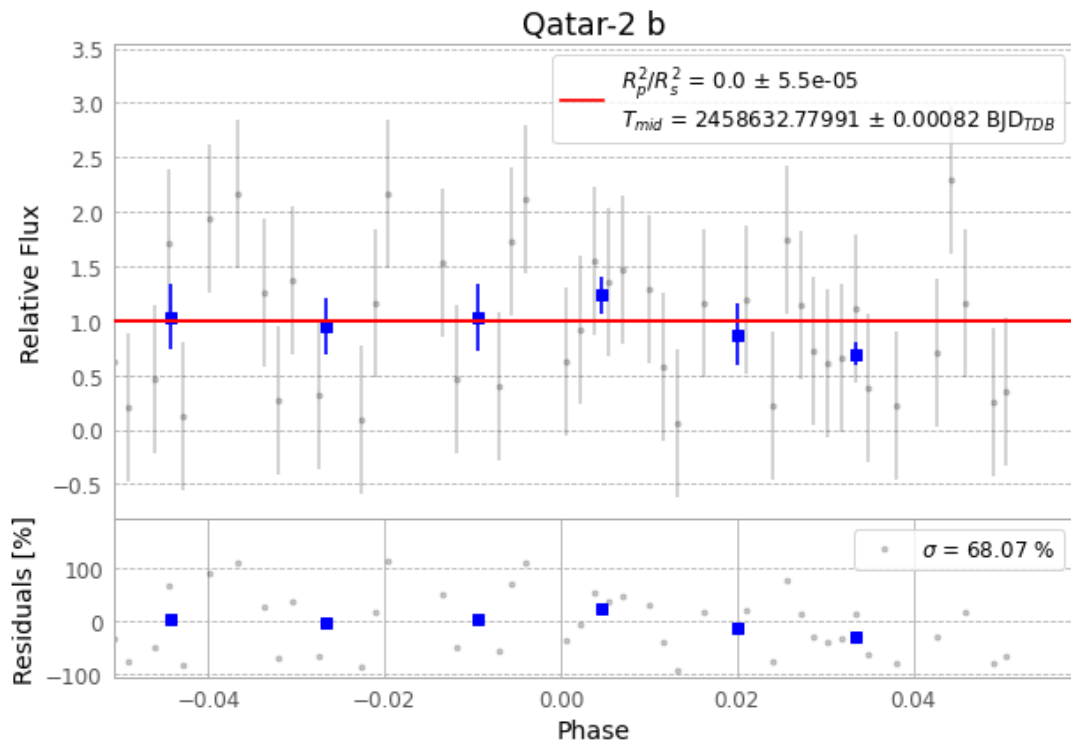


Figure 1 — FinalLightCurve_Qatar-2 b_2019-05-28.png (SHA-256 79c3b4c151e9b253...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [538, 439], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8dad6ad021a72157...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), Qatar - 2190529045412.FITS → Qatar - 2190529084212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-190529.log (072555a154b1...), FinalLightCurve_Qatar-2 b_2019-05-28.png (79c3b4c151e9...), FinalLightCurve_Qatar-2 b_2019-05-28.csv (6d580deba0f...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:17Z.